

ELIMINATING THE $4R - 7$ ALTERNATIVE FOR DIAMETER-TWO MIXED ALMOST MOORE GRAPHS

ABSTRACT. López and Miret derived two arithmetic alternatives for the parameters of a totally regular mixed almost Moore graph of diameter two. We prove that their alternative $c^2 = 4r - 7$ cannot occur when the undirected degree satisfies $r > 2$ and the directed degree satisfies $z \geq 1$. Consequently, in the standard simple mixed-graph model, every such graph with $r > 2$ and $z \geq 1$ must satisfy

$$c^2 = 4r + 1, \quad c \mid (4z + 1)(4z - 7)$$

for some odd integer c . The square condition is equivalent to $r = s(s+1)$ for an integer $s \geq 2$. In particular, no such graph has undirected degree $r = 4$, settling the first open row in the table of López and Miret.

1. INTRODUCTION

A mixed almost Moore graph of diameter two and degrees (r, z) has order

$$n = (r + z)^2 + z.$$

For totally regular graphs with even undirected degree $r > 2$ and directed degree $z \geq 1$, López and Miret proved that one of the following alternatives is necessary [2, Theorem 2]:

$$\begin{aligned} (1) \quad & c_1^2 = 4r + 1, & c_1 \mid (4z + 1)(4z - 7), \\ (2) \quad & c_2^2 = 4r - 7, & c_2 \mid (16z^2 + 40z - 23). \end{aligned}$$

Here c_1, c_2 are odd integers. The family $r = 4$, which lies in (2), was listed as unknown in [2, Table 1] and again in [1, Table 1].

In the proof of [2, Proposition 4], the cubic-trace computation is displayed for (1), while the other branch is said to follow by the same method. We carry out the branch-dependent calculation. The relevant difference of cubes is $c(r-1)$ in (2), rather than $c(r+1)$ as in (1); the resulting trace is negative, which is impossible.

2. THE TRACE OBSTRUCTION

Theorem 1. *There is no totally regular mixed almost Moore graph of diameter two with undirected degree $r > 2$, directed degree $z \geq 1$, and $4r - 7$ a square.*

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Proof. Let $d = r + z$ and

$$n = d^2 + d - r.$$

Let A be the adjacency matrix of the graph, let P be the permutation matrix of its repeat automorphism, and let J be the all-ones matrix. By [2, Eq. (1)],

$$(3) \quad I + A + A^2 = J + rI + P.$$

Moreover, total regularity and invariance under the repeat automorphism give

$$(4) \quad AJ = JA = dJ, \quad AP = PA.$$

Put

$$f = \text{tr } P, \quad t = \text{tr}(AP).$$

Since A and P are nonnegative integer matrices, both are nonnegative integers. Take traces in (3), and then multiply (3) by A and take traces again. Using $\text{tr } A = 0$, $\text{tr } J = n$, and $\text{tr}(AJ) = d \text{tr } J = dn$, the last by (4), this gives

$$(5) \quad \text{tr}(A^2) = rn + f, \quad \text{tr}(A^3) = zn - f + t.$$

Assume that $c^2 = 4r - 7$ with $c > 0$. Since $4r - 7$ is odd, so is c , whence $c^2 \equiv 1 \pmod{8}$ and r is even; together with $r > 2$ and $z \geq 1$ this places the graph under the hypotheses of [2, Theorem 2]. Moreover, the integer $4r + 1$ is not a square: otherwise, if $h^2 = 4r + 1$ with $h > 0$, then h is odd, again because $4r + 1$ is odd, and

$$(h - c)(h + c) = 8,$$

and since h and c are odd both factors are even, while $h^2 - c^2 = 8 > 0$ makes both positive, so $\{h - c, h + c\} = \{2, 4\}$. This forces $h = 3$, $c = 1$, and $r = 2$, contradicting $r > 2$. Thus the third case in the proof of [2, Theorem 2] applies.

Let q_j be the number of cycles of length j in P , and define

$$s_i = \sum_{i|j} q_j.$$

Euler's totient function φ and the Möbius function μ satisfy

$$(6) \quad \sum_{i=1}^n s_i \varphi(i) = n, \quad \sum_{i=1}^n s_i \mu(i) = q_1 = f.$$

Indeed, these follow by summing the identities

$$\sum_{i|j} \varphi(i) = j, \quad \sum_{i|j} \mu(i) = \mathbf{1}_{j=1}$$

over the cycles of P .

Set

$$\alpha = \frac{-1 + c}{2}, \quad \beta = \frac{-1 - c}{2}.$$

The characteristic-polynomial factorization in the third case of [2, proof of Theorem 2] is

$$(7) \quad \chi_A(x) = (x - d)(x^2 + x - r)^{(s_1-1)/2}(x - \alpha)^a(x - \beta)^b \\ \times \prod_{i=3}^n \Phi_i(x^2 + x - (r - 1))^{s_i/2},$$

where $a, b \in \mathbb{Z}_{\geq 0}$, $a + b = s_2$, and Φ_i is the i th cyclotomic polynomial. The branch enters at the two lowest indices. Since $\Phi_1(y) = y - 1$ and $\Phi_2(y) = y + 1$, the polynomials $\Phi_i(x^2 + x - (r - 1))$ for $i = 1$ and $i = 2$ are $x^2 + x - r$ and $x^2 + x - (r - 2)$, of discriminants $4r + 1$ and $4r - 7$. Under (2) it is the second that splits over $\mathbb{Z}$, its roots being α and β , while the first stays irreducible; under (1) the roles are exchanged. This is the source of the $c(r - 1)$ in (9) below, in place of the $c(r + 1)$ of the first branch.

Write $\Delta = a - b$. Taking the sum of the roots in (7) and using (6), we obtain

$$0 = \text{tr } A = d - \frac{s_1 - 1}{2} + \frac{-s_2 + c\Delta}{2} - \frac{1}{2} \sum_{i=3}^n s_i \varphi(i) = d + \frac{1 + c\Delta - n}{2}.$$

Hence

$$(8) \quad c\Delta = n - 2d - 1.$$

If u, v are the roots of $x^2 + x - r$, then

$$u^3 + v^3 = -1 - 3r.$$

Moreover,

$$(9) \quad \alpha^3 + \beta^3 = 5 - 3r, \quad \alpha^3 - \beta^3 = c(r - 1).$$

For a primitive i th root of unity ζ , let $\gamma_\zeta, \delta_\zeta$ be the roots of

$$x^2 + x - (r - 1) = \zeta.$$

Then

$$\gamma_\zeta^3 + \delta_\zeta^3 = 2 - 3r - 3\zeta.$$

Summing the cubes of the roots in (7), evaluating

$$2(a\alpha^3 + b\beta^3) = s_2(\alpha^3 + \beta^3) + \Delta(\alpha^3 - \beta^3)$$

by (9), and using (6), yields

$$(10) \quad 2(\text{tr}(A^3) - d^3) = (-1 - 3r)(s_1 - 1) + (5 - 3r)s_2 + (r - 1)c\Delta \\ + \sum_{i=3}^n s_i((2 - 3r)\varphi(i) - 3\mu(i)) \\ = (r - 1)c\Delta + (2 - 3r)n + 3r + 1 - 3f.$$

Comparing (5) and (10), and then using (8), $z = d - r$, and $n = d^2 + d - r$, gives

$$(11) \quad \begin{aligned} 2t &= 2d^3 - 2zn + (r - 1)(n - 2d - 1) + (2 - 3r)n + 3r + 1 - f \\ &= -d^2 + 3d + r + 2 - f. \end{aligned}$$

Recall that r is even; with $r > 2$ this gives $r \geq 4$, and $z \geq 1$ gives $d = r + z \geq r + 1$. As $f \geq 0$ and $-d^2 + 3d$ is decreasing for $d \geq 2$, (11) gives

$$2t \leq -(r + 1)^2 + 3(r + 1) + r + 2 = -r^2 + 2r + 4 < 0,$$

the final inequality because $r \geq 4$. This contradicts $t \geq 0$. $\square$

Corollary 2. *In the standard simple mixed-graph model, every mixed almost Moore graph of diameter two with $r > 2$ and $z \geq 1$ satisfies*

$$c^2 = 4r + 1, \quad c \mid (4z + 1)(4z - 7)$$

for some odd integer c . Equivalently, the square condition can be written as

$$r = s(s + 1)$$

for an integer $s \geq 2$. In particular, no such graph has $r = 4$, for any $z \geq 1$.

Proof. Mixed almost Moore graphs of diameter two are totally regular [4, Theorem 2.8], and their undirected degree is even [2, Proposition 1]. Thus the hypotheses of [2, Theorem 2] are met, $r > 2$ and $z \geq 1$ by assumption and the parity of r by the foregoing, and that theorem applies. Theorem 1 eliminates its second alternative, leaving the first. Writing the positive odd integer c as $2s + 1$ gives

$$r = \frac{c^2 - 1}{4} = s(s + 1).$$

For $r = 4$, the remaining alternative would require $c^2 = 17$, which is impossible. $\square$

3. STATUS AND SCOPE

Corollary 2 gives a partial negative answer to [2, Question 3] and to the broader diameter-two existence problem [3, Problem 5.2]. It settles the $r = 4$ family that was recorded as unknown in [2, Table 1] and [1, Table 1].

What is settled is one of the two arithmetic alternatives of [2, Theorem 2]: Theorem 1 refutes the branch $c^2 = 4r - 7$ for every $r > 2$ and every $z \geq 1$, uniformly in both degrees, so the $r = 4$ row is a special case of a general statement rather than a finite check. What is not settled is the branch $c^2 = 4r + 1$; for the degrees $r = s(s + 1)$ with $s \geq 2$ the existence question remains open. The hypotheses $r > 2$ and $z \geq 1$ are those of [2, Theorem 2], whose remaining hypothesis that r be even is automatic here, and the hypothesis $z \geq 1$ is used rather than decorative: substituting $z = 0$, that is $d = r$, in (11) gives $2t = -r^2 + 4r + 2 - f$, which at $r = 4$ is $2 - f$ and so permits $t \geq 0$. The argument therefore says nothing about $z = 0$, nor about $r \leq 2$.

The proof uses no computation: its decisive step is the evaluation of (11) at $d = r + 1$, which can be checked by hand. It is not self-contained. The matrix identity (3) is quoted from [2, Eq. (1)] and the characteristic-polynomial factorization (7) from [2, proof of Theorem 2], whose case analysis we do not reproduce; Corollary 2 further imports total regularity from [4, Theorem 2.8] and the parity of r from [2, Proposition 1]. What is carried out here is the branch-dependent cubic-trace calculation and the consequence drawn from it. We are not aware of prior work eliminating the $4r - 7$ branch; the literature we have located that cites [2], including [1], restates the $r = 4$ case as open.

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